

Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 5: Food Tests Practical (Part 2) — Detecting Proteins and Fats in the Kenyan Meal

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	D	D	B	D	C	C	A

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share prediction of protein-positive and fat-positive samples

Q1. Before doing any tests, a learner predicts which nutrients are found in a cup of milk from the Kenyan meal samples. Which prediction is correct?

- A. Milk contains protein only, with no fat
- B. Milk contains fat only, with no protein
- C. Milk contains both protein and fat**
- D. Milk contains neither protein nor fat because it is a liquid

Correct answer: C. Milk is a familiar source of both protein and fat, so it should give a positive Biuret test and a positive ethanol emulsion test. Many learners wrongly think milk has no fat.

USE AFTER: OBSERVE PHASE — Observe: Biuret test on the six labelled samples (NaOH first, then CuSO_4)

Q2. In the Biuret test for protein, a learner puts 2 cm³ of bean broth in a test tube. Which sequence of steps is correct?

- A. Add 4–5 drops of copper sulphate solution first, then add dilute NaOH
- B. Add absolute ethanol first, then pour into cold water
- C. Add iodine solution first, then heat the tube in a water bath
- D. Add dilute NaOH first, mix gently, then add 4–5 drops of 1% copper sulphate solution**

Correct answer: D. NaOH is added first and then copper sulphate. Adding the reagents in the wrong order gives a false blue result instead of violet.

USE AFTER: OBSERVE PHASE — Observe: recording Biuret results and the whole-class collated results table

Q3. During the Biuret test, a group records a violet colour in the tube containing nyama choma extract. What should they conclude?

- A. Reducing sugar is present in the nyama choma
- B. Fat is present in the nyama choma
- C. Starch is present in the nyama choma
- D. Protein is present in the nyama choma**

Correct answer: D. A violet or purple colour with Biuret reagent is the positive result for protein. Starch, fat and reducing sugars are detected by other tests.

USE AFTER: OBSERVE PHASE — Observe: ethanol emulsion test on butter, cooking oil and the other samples

Q4. In the ethanol emulsion test, a learner dissolves 2 cm³ of cooking oil in 5 cm³ of absolute ethanol. The learner then pours the solution into a test tube containing 5 cm³ of cold water. What result shows that fat is present?

- A. The mixture turns brick-red
- B. The mixture turns milky white as an emulsion forms**
- C. The mixture turns blue-black
- D. The mixture turns violet

Correct answer: B. Fat dissolves in ethanol but is insoluble in water. When the solution is poured into water, tiny fat droplets form a milky-white emulsion, which is a positive result.

USE AFTER: OBSERVE PHASE — Observe: teacher safety demonstration and bench safety monitor check for flames

Q5. A learner is about to add absolute ethanol to a test tube during the ethanol emulsion test. What safety step must be done first?

- A. Put on a wet cloth so it can be used to blow out a flame if needed
- B. Warm the ethanol gently over a flame so it dissolves the fat faster
- C. Smell the ethanol closely to check that it is the correct liquid
- D. Make sure no flame or spirit lamp is burning on the bench**

Correct answer: D. Ethanol is flammable, so all flames on the bench must be out before it is used. Heating it or sniffing it closely is unsafe.

USE AFTER: EXPLAIN PHASE — Explain: Question 1 – why nyama choma turns violet with Biuret reagent

Q6. Nyama choma gives a strong violet colour with Biuret reagent. Which explanation of this colour change is correct?

- A. Copper ions react with starch molecules to form a violet complex
- B. NaOH turns violet when it reacts with fat droplets
- C. Copper ions coordinate with the peptide bonds in protein chains, producing the violet colour**
- D. Ethanol dissolves the protein and turns violet in water

Correct answer: C. In the Biuret test, copper ions coordinate with the peptide bonds in protein chains. More peptide bonds give a stronger violet colour.

USE AFTER: EXPLAIN PHASE — Explain: Question 2 – link between protein in nyama choma and pepsin

Q7. Learners have shown that nyama choma is rich in protein. Which statement correctly links this to digestion in the stomach?

- A. Lipase in the stomach digests the protein into amino acids
- B. Amylase in the stomach digests the protein at neutral pH
- C. Pepsin, a protease, digests the protein, and the low pH of gastric acid activates it**
- D. Bile from the gall bladder digests the protein in the stomach

Correct answer: C. The stomach secretes pepsinogen, which is activated into pepsin by the acid (low pH). Pepsin is a protease that breaks protein into peptides.

USE AFTER: MODEL BUILDING PHASE — Model: annotating the cumulative alimentary canal diagram at the small intestine (bile salts + lipase)

Q8. Cooking oil gave a positive ethanol emulsion test. Which statement about its digestion in the small intestine is correct?

- A. Bile salts emulsify the fat, and then lipase digests it into fatty acids and glycerol**
- B. Fat is digested in the stomach by pepsin before it reaches the small intestine
- C. Lipase emulsifies the fat, and then bile salts digest it into fatty acids and glycerol
- D. Bile is an enzyme that chemically digests the fat into fatty acids and glycerol

Correct answer: A. Fat is insoluble in water, so bile salts must first emulsify it into small droplets. Lipase can then digest it into fatty acids and glycerol. Bile is not an enzyme.